

The empirical recurrence for OEIS A237062: recovering a conjecture that is not written down

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Abstract

OEIS A237062 records “Empirical recurrence of order 64”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 213$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 64, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 64 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237062 is “Number of $(n+1) \times (3+1)$ 0..2 arrays with the maximum plus the upper median plus the minimum minus the lower median of every 2×2 subblock equal.”. It has offset 1 and begins

826, 5034, 32336, 243256, 1812742, 15086720, 119493128, 1047778362, ...

The entry states:

Empirical recurrence of order 64 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 09:47:07, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 213$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 213$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 426$ exact terms of the model returns order 64 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{64} c_i a(n-i),$$

c_1	14	c_{17}	13898252323277	c_{33}	775311780943181959	c_{49}	82845882732048416
c_2	155	c_{18}	−33419242153815	c_{34}	−87900710266521689	c_{50}	41497774635647904
c_3	−2926	c_{19}	−111947122025002	c_{35}	−1212852787383485588	c_{51}	−22518734722941760
c_4	−6040	c_{20}	228917065443925	c_{36}	−18159285205931616	c_{52}	−12496183181743616
c_5	244521	c_{21}	733342909262495	c_{37}	1550269674991792532	c_{53}	4590009805465216
c_6	−164483	c_{22}	−1219480218179164	c_{38}	186721846976722302	c_{54}	2848783743085568
c_7	−11263921	c_{23}	−3910631636706057	c_{39}	−1617927889092342392	c_{55}	−666527588560896
c_8	21238672	c_{24}	5077215771414469	c_{40}	−336263932844312576	c_{56}	−476906525168640
c_9	335146840	c_{25}	16973727829173276	c_{41}	1375891660801136962	c_{57}	62453272052736
c_{10}	−838909982	c_{26}	−16513175556232755	c_{42}	386913316176477434	c_{58}	55743780907008
c_{11}	−7072996336	c_{27}	−59968942798729664	c_{43}	−949856475438318196	c_{59}	−2873895211008
c_{12}	19721506960	c_{28}	41635759309154095	c_{44}	−326804386366699776	c_{60}	−4163406077952
c_{13}	112388626043	c_{29}	172567185846981753	c_{45}	529142107884198200	c_{61}	−28607201280
c_{14}	−318347840079	c_{30}	−79677765087090516	c_{46}	211400822646920752	c_{62}	165801000960
c_{15}	−1397038560056	c_{31}	−404869034023921574	c_{47}	−235714568148881312	c_{63}	6589513728
c_{16}	3753859926320	c_{32}	109391191856311744	c_{48}	−106225582197826496	c_{64}	−2080899072

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 64, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $64 + 4$ terms removed returns order 64 as well, so $\deg q_{\min} = 64 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $64 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 64 that this sequence satisfies — and that family turns out to have exactly one member.

References

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